

Utility Functions for Optimization: A Decision

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Abstract—Bayesian optimization is usually taught as a catalogue of acquisition functions, which obscures the fact that most of them are one construction applied to different objects: a single step of lookahead on a utility function that scores the collected data. We present the utility functions of Chapter 6 of Garnett’s Bayesian Optimization in one notation, trace each to the algorithm it induces, and test the claims numerically on a campaign resource allocation problem modelled with a Gaussian process. Six standard utilities rank the same two datasets in incompatible orders, and one behaves opposite to the way the source predicts. The widely reported maximum observed value tracks the utility it approximates only above a signal-to-noise ratio of about three. The penalty for excessive optimism about uncertainty is roughly seven times the penalty for excessive caution. And the choice of utility decides not merely which of four policies ranks first but whether any difference is measurable at all: resolving the same top-two gap needs about three replicate runs under one utility and four hundred under another.

Index Terms—Bayesian optimization, decision theory, utility functions, Gaussian processes, acquisition functions, information gain

I. INTRODUCTION

Optimizing an expensive black-box function is a problem of acting under uncertainty. We hold beliefs about an objective f , we may query it a small number of times, and each query must be chosen before we know what it will return. Bayesian optimization addresses this by carrying a probabilistic model of f and using it to decide where to look next [1], [5].

The model, however, does not tell us what to do. It tells us what we believe. A Gaussian process posterior is a statement about the objective; it contains no statement about which of two datasets constitutes a better outcome. That judgement has to be supplied separately, and it is supplied by a *utility function* $u(\mathcal{D})$, which assigns a score to the data returned by an optimization run. Once beliefs and preferences are both specified, Bayesian decision theory determines the optimal policy mechanically: at every step, query the point that maximizes the expected utility of the final dataset [1].

This framing is powerful because it turns a proliferation of heuristics into a single mechanism applied to different inputs. Expected Improvement [3], the Knowledge Gradient [4], GP-UCB [1] and the entropy search family [1, §7.6]

The author selected the topic, developed the campaign case study and specified the direction of the experiments. A large language model was used to write the accompanying Gaussian process implementation and to generate the figures.

are not independent inventions. Each is tied to a different answer to one question: what would make this optimization run a success? For most of them the tie is exact, one step of lookahead on the corresponding utility; for GP-UCB it is looser, the utility supplying the notion of regret against which the algorithm is analysed.

The framing has a consequence that motivates the experimental part of this paper. The utility function is not only what a policy is derived from; it is also the only well-defined notion of how good a returned dataset is, and so it is simultaneously the design choice and the evaluation metric. If two authors derive policies from two different utilities and compare them, each is scored on the other’s terms, and the comparison says more about the yardstick than about the algorithms.

This paper does three things. It presents the utility functions of [1, ch. 6] in one notation, tracing each to the acquisition function it induces. It instantiates them in a concrete decision problem, campaign resource allocation, with a fully specified Gaussian process model. And it tests the source’s claims numerically, confirming most of them, qualifying one, and supplying a number where the source leaves a threshold unspecified.

II. BACKGROUND AND PROBLEM SETTING

A. Objective, model and data

We seek to maximize an objective $f : \mathcal{X} \rightarrow \mathbb{R}$ over a domain $\mathcal{X}$. Observations are noisy: querying at x returns $y = f(x) + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, \sigma_n^2)$. After t queries we hold a dataset $\mathcal{D} = (\mathbf{x}, \mathbf{y})$.

We model f with a Gaussian process, so that the posterior at any x is Gaussian with mean $\mu_{\mathcal{D}}(x)$ and standard deviation $\sigma_{\mathcal{D}}(x)$ available in closed form [2]. Following [1, ch. 6] we write $\phi = f(x)$ for the objective value at a point under consideration, and $\mu_{\mathcal{D}}(x) = \mathbb{E}[\phi \mid x, \mathcal{D}]$. The utility functions below need only a posterior mean to exist; we use a Gaussian process because it makes every quantity computable.

B. Optimization as a sequential decision problem

Bayesian decision theory prescribes an optimal policy once two ingredients are fixed: a model, expressing beliefs, and a utility function $u(\mathcal{D})$, expressing preferences over the data an optimization run might return [1]. The optimal action at each step maximizes the expected utility of the terminal dataset; computing this exactly costs exponentially in the remaining

budget, so one truncates the lookahead, and truncating to a single step is what produces the standard acquisition functions [1, §5.3]. This paper concerns the second ingredient throughout: we take the model as given and ask what u should be.

C. The terminal recommendation

A natural way to build a utility function is to ask what the data are *for*. In many applications, optimization is followed by a decision: we tune hyperparameters and then deploy one configuration; we test doses and then prescribe one; we poll districts and then commit our remaining resources to one. The data matter only insofar as they support that final choice well [1, §6.1].

Formalizing this requires two objects. First, an *action space* $\mathcal{A} \subset \mathcal{X}$ of admissible recommendations. Second, a *terminal utility* $v(\phi)$ scoring a recommendation in hindsight by the objective value it turned out to have. Given these, the rational recommendation maximizes expected terminal utility, $x \in \arg \max_{x' \in \mathcal{A}} \mathbb{E}[v(\phi') \mid x', \mathcal{D}]$, and the value of the data is the expected utility that this optimal recommendation achieves:

$$u(\mathcal{D}) = \max_{x' \in \mathcal{A}} \mathbb{E}[v(\phi') \mid x', \mathcal{D}]. \quad (1)$$

Equation (1) is (6.1) of [1] and is the central construction of this paper. It collapses a two-stage problem, collect data then recommend a point, into a single score on the data alone: the maximum and the expectation together remove the dependence on both the recommended point and its unknown value. Every utility in Section IV that concerns terminal recommendation is a specialization of (1) obtained by choosing $\mathcal{A}$ and v .

III. THE TWO DESIGN CHOICES

A. The action space

The action space answers a question that sounds procedural but is substantive: after optimization, where are we permitted to point?

The conservative extreme is to restrict $\mathcal{A}$ to the visited locations $\mathbf{x}$, so that every candidate has at least one real observation behind it. The permissive extreme is $\mathcal{A} = \mathcal{X}$: we may recommend anywhere, including points we never evaluated, if the model says they are good. As [1, §6.1] puts it, recommending an unvisited point for perpetual use is ultimately a question of faith in the model's beliefs.

Intermediate options exist, such as allowing only points whose value is known to some tolerance. But the two extremes are what matter here: this single choice, and nothing else, separates the two most widely used utilities in Bayesian optimization.

B. Risk tolerance

The second choice is the shape of v . Since we are maximizing, v must be increasing in ϕ , but its curvature is free, and the curvature encodes an attitude toward uncertainty.

The simplest choice is linear, $v(\phi) = \phi$, giving $\mathbb{E}[v(\phi) \mid x, \mathcal{D}] = \mu_{\mathcal{D}}(x)$. The optimal recommendation then maximizes the posterior mean and ignores $\sigma_{\mathcal{D}}$ entirely: two points with equal expected value are equally attractive however differently

uncertain they are. This is *risk neutrality*, and it is the standard assumption in the Bayesian optimization literature.

Risk neutrality is an assumption rather than a default. The useful notion for seeing this is the *certainty equivalent*: the risk-free value ϕ' for which we would be indifferent between a guaranteed ϕ' and the uncertain prospect at x , defined by $v(\phi') = \mathbb{E}[v(\phi) \mid x, \mathcal{D}]$. Under a concave v , Jensen's inequality gives

$$v(\phi') = \mathbb{E}[v(\phi) \mid x, \mathcal{D}] \leq v(\mathbb{E}[\phi \mid x, \mathcal{D}]) = v(\mu_{\mathcal{D}}(x)), \quad (2)$$

so the certainty equivalent of an uncertain recommendation falls below its expected value: we would trade it for a safe alternative worth less on average. That is risk aversion. A convex v reverses the inequality and encodes risk seeking.

A tractable parametric family combines a point's expected reward and its risk directly,

$$\mu_{\mathcal{D}}(x) + \beta \sigma_{\mathcal{D}}(x), \quad (3)$$

with $\beta < 0$ risk averse, $\beta = 0$ risk neutral and $\beta > 0$ risk seeking. Because μ and σ share units, the combination is dimensionally sensible. The source notes that risk-averse utilities are rarely encountered in Bayesian optimization while arguing they may be preferable in practice [1, §6.1]; Section VI-C measures the trade-off that (3) makes.

IV. A CATALOGUE OF UTILITY FUNCTIONS

For ease of checking against the source, our (4), (5), (6), (7) and (8) are equations (6.3), (6.5), (6.6), (6.7) and (6.8) of [1] respectively.

A. Simple reward

Combining risk neutrality with the conservative action space $\mathcal{A} = \mathbf{x}$ in (1) gives the *simple reward*:

$$u(\mathcal{D}) = \max \mu_{\mathcal{D}}(\mathbf{x}), \quad (4)$$

the largest posterior mean among the visited locations. When observations are exact this reduces to $u(\mathcal{D}) = \max \phi$, the best value actually seen [1, (6.4)].

One step of lookahead on (4) yields Expected Improvement [3], the most widely deployed acquisition function in practice. Simple reward depends on the posterior only *at the observed locations*: it is a local measure, which is both its virtue and its limit.

B. Global reward

Keeping risk neutrality but taking $\mathcal{A} = \mathcal{X}$ gives the *global reward*:

$$u(\mathcal{D}) = \max_{x \in \mathcal{X}} \mu_{\mathcal{D}}(x), \quad (5)$$

the global maximum of the posterior mean. The recommendation may now fall anywhere, including in regions never evaluated, provided the model predicts a high value there. One step of lookahead yields the Knowledge Gradient [4].

The contrast with simple reward is best stated as a contrast of attitudes: simple reward expresses local safety, global reward expresses model-trusting ambition. Neither dominates. Global reward extracts more from a good model and is punished harder by a bad one.

C. A tempting but incoherent alternative

A quantity that appears in the literature, and that resembles simple reward closely enough to be mistaken for it, is the maximum raw observation:

$$u(\mathcal{D}) \stackrel{?}{=} \max \mathbf{y}. \quad (6)$$

When observations are exact, (6) coincides with simple reward. When they are noisy it does not, because $\max \mathbf{y}$ scores the measurement rather than the objective: a dataset with one lucky outlier beats a dataset of consistently good, accurately measured values. The preference encoded is a preference for measurement accidents over optimization progress [1, §6.1]. The posterior mean shrinks a noisy observation toward what its neighbours and the prior imply, and so discounts the outlier automatically; $\max \mathbf{y}$ has no such mechanism. The source calls (6) a reasonable *approximation* when the signal-to-noise ratio is high, without saying how high. Section VI-B supplies a number.

D. Cumulative reward

All of the above assume only the best point matters. Sometimes the optimization process itself has consequences: if each query is an action in a live system, a bad query is a real cost, not merely a wasted evaluation. The *cumulative reward* scores the average rather than the maximum:

$$u(\mathcal{D}) = \sum_i y_i. \quad (7)$$

This is the utility underlying the bandit view of optimization and the regret analyses that accompany it [1]. It is also the natural objective of active search, where one seeks as many members of a rare valuable class as possible and every find counts equally. Note that (7) depends on the data alone and not on the model.

E. Information gain

The final utility abandons the premise that success means finding high values, and asks instead how much was *learned*. Choose a random variable ω of interest; the natural choices in optimization are the location x^* of the global optimum or its value f^* . Measure the reduction in differential entropy achieved by the data:

$$u(\mathcal{D}) = H[\omega] - H[\omega | \mathcal{D}]. \quad (8)$$

An alternative definition uses the Kullback–Leibler divergence between posterior and prior, which is always nonnegative whereas (8) can be negative for surprising data. The two have equal expectation with respect to observed values and induce identical policies [1, §6.3], so the choice is a matter of taste.

One step of lookahead gives mutual information, the basis of the entropy search family: Entropy Search [1, §7.6] takes $\omega = x^*$, while Max-value Entropy Search [1, §7.6] takes $\omega = f^*$ and is markedly cheaper because f^* is scalar. Unlike the reward utilities, (8) is not intrinsically tied to optimization: choosing ω freely turns it into a general formulation of experimental design, and, arguably, of scientific inquiry.

TABLE I

EACH UTILITY FUNCTION, THE DESIGN CHOICES THAT PRODUCE IT, AND THE ALGORITHM IT CORRESPONDS TO. THE CORRESPONDENCE IS ONE-STEP LOOKAHEAD EXCEPT IN THE CUMULATIVE-REWARD ROW, WHERE THE UTILITY INSTEAD SUPPLIES THE REGRET NOTION USED TO ANALYSE THE ALGORITHMS LISTED.

Utility	Action space	Acquisition	Ref.
Simple reward (4)	visited $\mathbf{x}$	Expected Improvement	[3]
Global reward (5)	domain $\mathcal{X}$	Knowledge Gradient	[4]
Cumulative reward (7)	—	GP-UCB, active search	[1]
Info. gain, $\omega = x^*$ (8)	—	Entropy Search	[1]
Info. gain, $\omega = f^*$ (8)	—	Max-value ES	[1]
$\max \mathbf{y}$ (6)	visited $\mathbf{x}$	(approximates EI)	[1]

F. Dependence on the model

Simple and global reward are defined through $\mu_{\mathcal{D}}$, and information gain through the posterior belief about x^* or f^* . All are therefore functions of the model as much as of the data: a dataset’s score changes if the model changes, with no new observations. Model averaging mitigates this [1, §4.4]. Model-agnostic utilities exist but are limited to cumulative reward and $\max \mathbf{y}$, which is precisely why the latter is tempting and, as Section IV-C argues, why it fails.

G. Summary of the mapping

Table I collects the correspondence between utilities, their design choices, and the algorithms they induce.

V. CASE STUDY: CAMPAIGN ALLOCATION

A. The decision problem

To make the framework concrete we use a resource allocation problem with the structure of a terminal recommendation. A campaign has one remaining day of effort and must commit it to a single electoral district. Districts are indexed by a scalar coordinate $x \in [0, 1]$ summarizing their demographic composition, so that demographically similar districts are nearby. The objective $f(x)$ is the shift in projected vote margin, in percentage points, that concentrating the final day’s effort on district x would produce. Polling a district returns a noisy estimate of f at cost, which is what makes the problem one of expensive black-box optimization.

Each design choice is interpretable here. The action space is the question of whether the campaign may commit to a district it never polled. Risk tolerance is the question of whether a campaign that is behind should gamble. Cumulative reward applies if the polls themselves move opinion; information gain applies if the goal is to understand the electorate for a later decision rather than to act now.

B. Model and implementation

We place a zero-mean Gaussian process prior on f with a squared exponential covariance,

$$k(x, x') = \sigma_f^2 \exp\left(-\frac{1}{2} \left(\frac{x-x'}{\ell}\right)^2\right), \quad (9)$$

with amplitude $\sigma_f = 3$ points, length scale $\ell = 0.08$, and observation noise $\sigma_n = 0.5$ points unless stated otherwise. A

TABLE II
SIX UTILITIES, TWO DATASETS, AND NO CONSENSUS. INFORMATION GAIN IS A MONTE CARLO ESTIMATE OVER 6×10^4 POSTERIOR SAMPLE PATHS, REPORTED WITH THE STANDARD ERROR OVER EIGHT INDEPENDENT REPETITIONS.

Utility	Dataset A	Dataset B	Prefers
Simple reward (4)	-0.98	+0.97	B
Global reward (5)	+1.52	+0.97	A
max y (6)	-1.00	+1.00	B
Cumulative reward (7)	-15.00	+1.00	B
Info. gain about x^* (nats)	0.878 ± 0.003	0.033 ± 0.003	A
Info. gain about f^* (nats)	0.004 ± 0.002	0.062 ± 0.002	B

zero prior mean encodes the assumption that a district is as likely to be unresponsive as responsive before any polling. The domain is discretized to a uniform grid of 501 points, which is fine relative to ℓ .

All Gaussian process conditioning, all six utility functionals and all acquisition functions were implemented directly in NumPy rather than taken from a library, so that the definitions used are exactly those of Section IV. Information gain is estimated by Monte Carlo: we draw 6×10^4 posterior sample paths on the grid, take x^* as the grid argmax, estimate $H[x^*]$ as a discrete entropy over grid cells, and estimate $H[f^*]$ from the sampled maxima with the Vasicek spacing estimator. Because information gain is a difference of entropies computed on the same grid, the discretization constant cancels. The Knowledge Gradient uses the exact expectation of a maximum of affine functions of a standard normal via the epigraph algorithm of [4], verified against a 3×10^6 -sample Monte Carlo estimate on 30 random instances (worst discrepancy 1.7 standard errors).

VI. EXPERIMENTS

A. The same data, five different verdicts

Section 6.5 of [1] argues qualitatively that these utilities can disagree, using two schematic figures and no numbers. We reproduce the argument quantitatively.

Dataset A consists of five polls, at $x = 0.10, 0.22, 0.40, 0.46, 0.52$, returning $-1.0, -1.6, -6.0, -5.4, -1.0$ points. Every value is below the prior mean: on its face, five straight losses. Dataset B consists of a single poll at $x = 0.50$ returning $+1.0$ points, a mildly encouraging result. Figure 1 shows the resulting posteriors, the recommendations induced by simple and global reward, and the induced belief $p(x^* | \mathcal{D})$.

Table II gives all six utility values for both datasets. The utilities split four against two, and the split is informative: simple reward, max y, cumulative reward and information gain about f^* all prefer B, while global reward and information gain about x^* prefer A. The dividing line is not reward-versus-information but local-versus-global, exactly the distinction [1, §6.5] draws. Two datasets, six defensible definitions of success, and a four-to-two split with a 2.5-point spread between the extremes.

First, the magnitude is large. Dataset A’s simple reward is negative while its global reward is $+1.52$ points, a swing of 2.5 points produced by nothing but the choice of action space. The analyst who will only recommend a polled district reports that the best option loses by a point; the strategist who will recommend anywhere reports a projected win by one and a half. Same five polls.

Second, information gain about x^* separates the two datasets by more than an order of magnitude, 0.878 against 0.033 nats. Dataset A’s steep implied gradient is highly informative about *where* the optimum is even though it contains no good values, which is exactly the phenomenon [1, §6.5] describes.

Third, and contrary to the source, information gain about f^* runs the other way: dataset B is more informative about the optimum’s *value* (0.062 against 0.004 nats), a gap far outside Monte Carlo error. Dataset A localizes the optimum to a narrow region, but the posterior standard deviation at the implied peak is still 2.08 points, so the value there remains almost as uncertain as a priori. Dataset B, by observing a value above the prior mean, truncates the lower tail of $p(f^*)$, learning a little about the value and nothing about the location. We therefore qualify the source’s claim: the two components of information gain need not move together, which is the practical reason Entropy Search and Max-value Entropy Search can behave differently on the same problem.

B. When is max y safe?

The left and centre panels of Figure 2 construct the reversal described in Section IV-C. The noisy scenario polls a genuinely flat objective twelve times with $\sigma_n = 3$ points, and one poll returns $+6.2$ points by chance. The clean scenario polls a real 2.6-point effect twelve times with $\sigma_n = 0.15$. By max y, the noisy dataset scores 6.2 against 2.80 and wins comfortably. By simple reward it scores 1.74 against 2.80 and loses; by global reward, 1.75 against 2.80. The posterior mean has shrunk the outlier by more than a factor of three, because its eleven neighbours contradict it. The campaign reading is direct: max y would send the candidate to a district whose apparent six-point swing is a polling artefact, in preference to a district with a real and accurately measured two-and-a-half-point swing.

The source observes that max y is an acceptable approximation when the signal-to-noise ratio is high, and leaves the threshold unspecified. We estimate it. For each noise level we generated 400 pairs of eight-observation datasets from the prior and recorded how often max y and simple reward ordered the pair identically. The right panel of Figure 2 shows the result. Agreement is 78% at $\sigma_f/\sigma_n = 0.5$, rises through 89% at unit signal-to-noise, and first reaches the 95% level at $\sigma_f/\sigma_n \approx 3$, reaching 100% by $\sigma_f/\sigma_n = 20$.

So max y is defensible only when the noise standard deviation is below roughly a third of the signal amplitude.

C. The price of risk aversion

Risk tolerance is the axis [1, §6.1] identifies as under-explored. We sweep β in (3) from -3 to $+3$ over 400 replicate

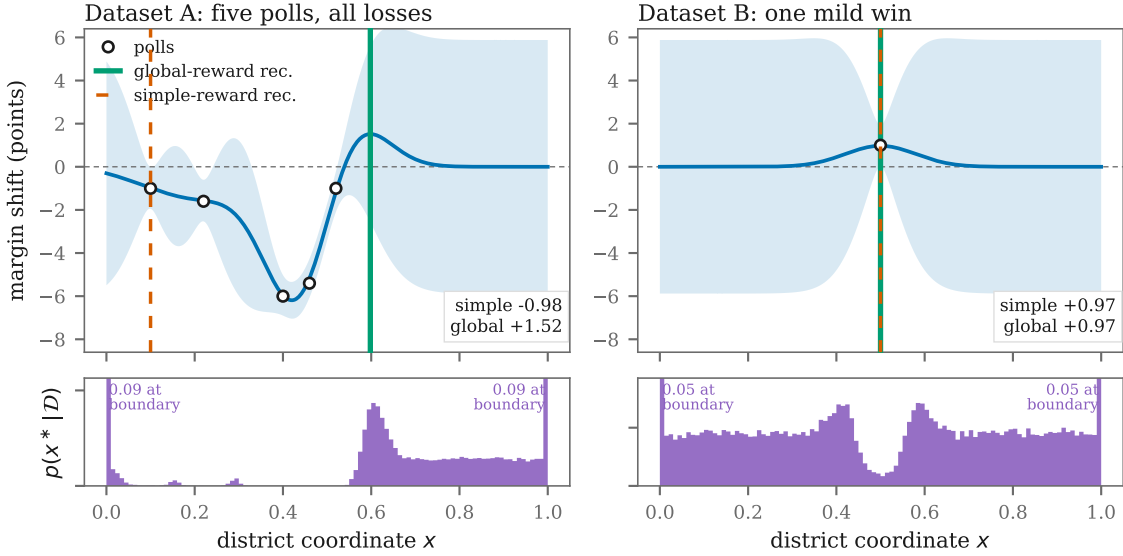


Fig. 1. Two polling outcomes and the recommendations they induce. Dataset A (left) contains five polls, every one of them below the prior mean, but the steep implied gradient pushes the posterior mean to +1.52 points in an unpollled region; the belief about the optimum’s location $p(x^* | \mathcal{D})$ concentrates sharply. Dataset B (right) contains one mildly good poll: the best value seen is higher, but the model has learned almost nothing about where the optimum lies. Simple reward prefers B; global reward and information gain about x^* prefer A.

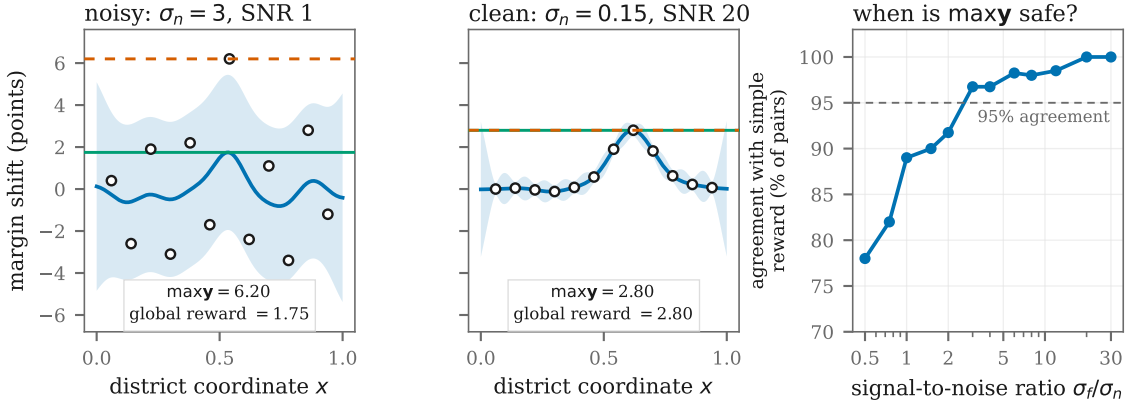


Fig. 2. The maxy pathology and its boundary. Left: twelve polls of a genuinely flat objective, taken with $\sigma_n = 3$; the apparent 6.2-point winner is noise, and the posterior mean correctly discounts it to 1.74. Centre: twelve accurate polls of a real 2.6-point effect, which every principled utility scores higher. maxy ranks these backwards. Right: over 400 random dataset pairs per noise level, maxy orders pairs the same way simple reward does at least 95% of the time only once the signal-to-noise ratio exceeds about three.

objectives drawn from the prior, each observed at six locations, and record the *true* objective value at the recommended district, which is available because the ground truth is synthetic.

What β controls is unambiguous: the posterior standard deviation at the recommended district rises monotonically from 0.48 points at $\beta = -2$ to 2.43 at $\beta = +2$, a five-fold range. The campaign can dial how much it is guessing.

What this purchases is strikingly asymmetric. Moving from neutral to averse costs 0.20 points of expected margin, about 6%; moving from neutral to seeking costs 1.52 points, about 45%. The penalty for excessive optimism is roughly seven times that for excessive caution. The tenth percentile falls from 0.68 under risk neutrality to -2.24 at $\beta = +2$, so a risk-seeking campaign has a one-in-ten chance of losing more than

two points. The mean is maximized at $\beta^* = -0.25$.

Risk aversion does not *improve* the realized distribution, and the reason is worth stating. Our ground truth is drawn from the same prior used for inference, so the posterior standard deviation is an honest statement of what is unknown, and the risk-neutral recommendation is by construction optimal in expectation. Any $\beta \neq 0$ can only trade expected value away. What risk aversion buys is a more *predictable* outcome, which is worth paying for when preferences are nonlinear in the objective, as a concave v in (2) encodes. The case where risk aversion should actually improve outcomes is the misspecified one, which our setup deliberately excludes.

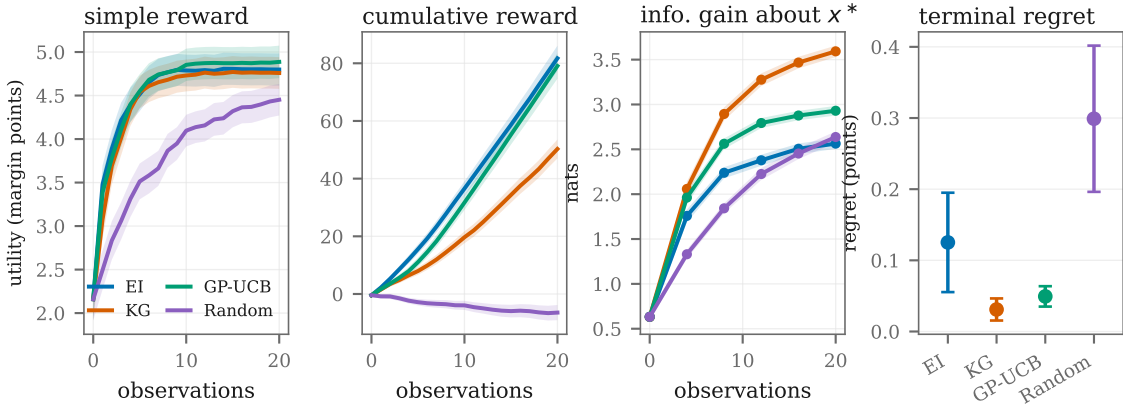


Fig. 3. Four policies run on the campaign problem for 20 observations beyond a common three-point initialization, over 120 replicate objectives drawn from the prior; bands and intervals are one standard error and 95% confidence respectively. Each panel scores the *same* runs by a different utility. Simple reward barely separates the three informed policies; cumulative reward and information gain separate them decisively and in opposite directions, with Expected Improvement first under one and last under the other.

TABLE III

TERMINAL UTILITIES AFTER 20 OBSERVATIONS, MEAN $\pm$ STANDARD ERROR OVER 120 REPLICATES. EACH COLUMN IS A DIFFERENT DEFINITION OF SUCCESS. REGRET IS $f^* - f(x_{\text{REC}})$ AT THE GLOBAL-REWARD RECOMMENDATION, SO LOWER IS BETTER. BEST ENTRY PER COLUMN IN BOLD.

Policy	Simple	Global	Cumul.	IG x^*	Regret
EI	4.80 $\pm$ 0.18	4.80 $\pm$ 0.18	81.7 $\pm$ 4.40	2.56 $\pm$ 0.06	0.13 $\pm$ 0.04
KG	4.76 $\pm$ 0.18	4.93 $\pm$ 0.18	50.3 $\pm$ 3.28	3.59 $\pm$ 0.06	0.03 $\pm$ 0.01
GP-UCB	4.89 $\pm$ 0.19	4.89 $\pm$ 0.19	79.1 $\pm$ 4.41	2.93 $\pm$ 0.05	0.05 $\pm$ 0.01
Random	4.45 $\pm$ 0.18	4.63 $\pm$ 0.19	-6.5 $\pm$ 2.64	2.64 $\pm$ 0.06	0.30 $\pm$ 0.05

TABLE IV

PAIRED COMPARISON OF THE BEST AND SECOND-BEST POLICY UNDER EACH UTILITY, AND THE NUMBER OF REPLICATE RUNS REQUIRED TO DETECT THAT SAME GAP WITH 80% POWER AT $\alpha = 0.05$. IDENTICAL RUNS; ONLY THE DEFINITION OF SUCCESS DIFFERS.

Utility	Best vs. 2nd	Gap	t	Reps needed
Simple reward	GP-UCB vs. EI	0.088	2.31	177
Global reward	KG vs. GP-UCB	0.041	1.55	391
Cumulative reward	EI vs. GP-UCB	2.652	3.58	74
Info. gain, x^*	KG vs. GP-UCB	0.663	16.95	3
Info. gain, f^*	GP-UCB vs. EI	0.164	8.91	12
Regret	KG vs. GP-UCB	0.018	1.95	246

D. The utility decides the comparison

The preceding experiments compare datasets; this one compares policies. We ran four on the campaign problem: Expected Improvement, from simple reward; the Knowledge Gradient, from global reward; GP-UCB with $\beta = 2$, motivated by cumulative reward; and uniform random search as a control. Each was given the same three-point initialization and 20 further observations, repeated over 120 objectives drawn from the prior, so the model is exactly the generating process. We then scored every run by *every* utility.

Figure 3 shows the trajectories and Table III the terminal values. Because all four policies see the same objective within a replicate, we compare them with paired differences, which removes the between-objective variance and is considerably more powerful than comparing unpaired means.

a) The metric decides whether any difference is visible:

The four policies are statistically indistinguishable under some utilities and enormously separated under others, on identical runs. Table IV asks, for each utility, how many replicates would be needed to resolve the best-versus-second gap at conventional power. The answer ranges over two orders of magnitude.

Under information gain about x^* the gap between the best and second-best policy is 17 standard errors wide and three replicate runs would suffice to establish it. Under simple

reward, the utility closest to what is routinely reported in applied work, the corresponding gap is 2.3 standard errors at 120 replicates, and reaching conventional power on it would take about 180 runs; under global reward, about 390. The same four policies, the same 120 runs, and a hundred-fold difference in the experimental effort needed to tell them apart. An experimenter who evaluates by simple reward and reports that two competent policies are indistinguishable may be describing the metric rather than the policies.

b) *Rank inversions are systematic, not noise:* The orderings do not merely blur; they reverse, and they reverse in the direction the utilities predict. The Knowledge Gradient, derived from global reward and therefore willing to spend queries anywhere in the domain, is first under global reward, first by a wide margin under information gain about x^* , and first on terminal regret, while accumulating a cumulative reward 38% below Expected Improvement's. Expected Improvement, derived from simple reward and consequently concentrating its queries near its incumbent, inverts this exactly: first under cumulative reward, and last of all four policies under information gain about x^* .

That last entry is worth stating plainly. Paired against random search, Expected Improvement's information gain about x^* is -0.07 ± 0.06 nats: after twenty observations it has

learned no more about *where* the optimum lies than uniform random sampling has. This is not a defect, as EI is designed to stand on a high observed value rather than to locate the optimum, and by that measure it beats random search by 0.35 ± 0.07 points. But the exploitative behaviour induced by simple reward carries a measurable informational cost.

c) What this does and does not show: It is tempting to summarize this as “each policy wins under its own utility,” and that would be too neat. Expected Improvement is not the best policy under simple reward in our runs; GP-UCB is, and the Knowledge Gradient’s lead under global reward is not significant. The defensible claim is weaker: the utility a policy is derived from predicts the *character* of its behaviour, exploitative or exploratory, and so predicts which metrics will flatter it, while simple and global reward often lack the resolution to separate competent policies at realistic replicate counts.

VII. DISCUSSION

A. What the experiments imply for practice

The utility should be chosen from the downstream decision, not from convention. Whether the recommendation may fall on an unevaluated point is a question about the application, not the algorithm: a campaign can commit to an unpolled district, a drug trial cannot prescribe an untested dose. That single choice moves the reported quality of a dataset by 2.5 points on a 3-point scale.

Reporting should state the utility, and should report the one that matches the application. Section VI-D shows that the scoring rule determines the statistical power of a comparison, not just its outcome: an author who finds no significant difference under simple reward may simply have chosen a metric needing three hundred replicate runs to see it.

Risk neutrality should be a decision rather than an inheritance, but the reason to depart from it is preference, not performance. The practical finding is the asymmetry: caution cost 6% of expected margin, optimism 45%. Where β must be guessed, guess low.

Finally, $\max_y$ should be retired as a reported metric except where the signal-to-noise ratio is documented and high. Simple reward costs one extra posterior evaluation and is correct by construction.

B. Limitations

The study is deliberately narrow so that every quantity is exactly computable. The domain is one-dimensional, and the disagreements may grow or shrink in higher dimensions. The model is well specified, since ground truth comes from the prior used for inference; this favours global reward and the Knowledge Gradient, which trust the model off-data, and misspecification should shift the balance toward simple reward and Expected Improvement. The objective is synthetic, so the campaign framing is an interpretive device rather than a validated model of electoral response. All policies use one-step lookahead.

VIII. CONCLUSION

Utility functions are not a preliminary to Bayesian optimization; they are where its content lies. The model expresses what we believe, the utility what we want, and only the pair determines an action. Working through Chapter 6 of [1] in one notation makes the standard acquisition functions look less like a catalogue of heuristics and more like one construction, Equation (1), applied to different answers to the question of what success means.

Our experiments show the choice is consequential rather than cosmetic. Six utilities ranked two datasets in incompatible orders, and information gain about the optimum’s value moved opposite to information gain about its location. The maximum observed value tracks the utility it approximates only above a signal-to-noise ratio of about three. Risk seeking proved about seven times as costly as risk aversion. And the choice of utility decided not only the ranking of four policies but whether a ranking was measurable at all: the utility is what one is measured by, and it sets the resolution of the measurement.

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